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T11 – 05

Assignment No. 1

Aim: Study of different sensors, controllers and wireless modules.

Sensor:

A sensor is defined as a device or a module that helps to detect any changes in physical quantity like pressure, force, or electrical quantity like current or any other form of energy.

1] Description of various Sensors are as follows:

1.1] IR Sensor (Infrared Sensor):

When we look into electromagnetic spectrum, infrared region is divided into three regions as Near Infrared, mid Infrared and far Infrared region. Infrared spectrum has a higher frequency range than microwave and lesser frequency than visible light. An infrared sensor is used for emitting and detecting IR radiation. By this principle, IR sensor can be used as Obstacle detector. There are two types of IR sensors as Active and Passive IR sensors.

- Passive IR sensor: When the sensor does not use any IR source to detect the emitted energy from the obstacles it acts as a Passive IR sensor. Examples like thermocouple, pyro electric detector and bolometers comes under passive sensors.
- Active IR sensor: When are there two components which acts as IR source and IR detector it is called as Active sensor. LED or laser diode act as IR source. Photodiode or phototransistors acts as IR detector.



1.2] Temperature & Thermocouple Sensors:

The analog sensor produces signals that are continuously varying with time. The output value from the sensor will be very small in range of microvolts or

milli volts. Due to this, signal-conditioning circuits are required for amplification. Analog to digital (ADC) converters are used to convert the analog signal obtained into a digital value.

Temperature sensor sense the temperature and measures the changes in the temperature. Other types of temperature sensors are [Thermocouples](#), [Thermistors](#), [Resistive Temperature Devices \(RTD\)](#) and Temperature Sensor ICs (LM35) etc.



1.3] Proximity Sensor:

Proximity sensor is a type of non-contact sensor used for object detection. It does not have any physical contact with the object. The object whose distance is to be measured is known as target. An IR light or electromagnetic radiation is used in a proximity sensor. There are different types of proximity [sensors like Inductive, capacitive](#), Ultrasonic, etc. applications: Object detection, to measure velocity, rotation identification, material detection, Reverse parking sensor, object counting.



1.4] Ultrasonic Sensor:

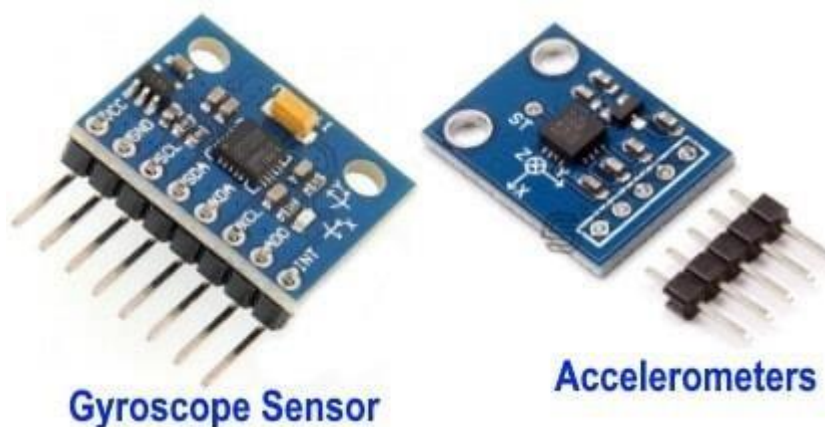
Ultrasonic sensors are used to measure distance or travel time using ultrasonic waves. A source will be used to emit ultrasonic wave. After wave hits the target, the waves are reflected and the detector collects the signal. The travel time between the transmitted wave and reflected wave is measured using Ultrasonic sensor. Optical sensors use two different element for transmitter and receiver. Whereas ultrasonic sensor uses single element for transmission and reception.



1.5] Accelerometers & Gyroscope Sensor:

Accelerometer is a type of sensor that is used to detect changes in position, velocity, and vibration by sensing motion. It can be either analog or digital type. In analog accelerometer, depending on volume of acceleration applied to accelerometer, continuous analog voltage signal is produced.

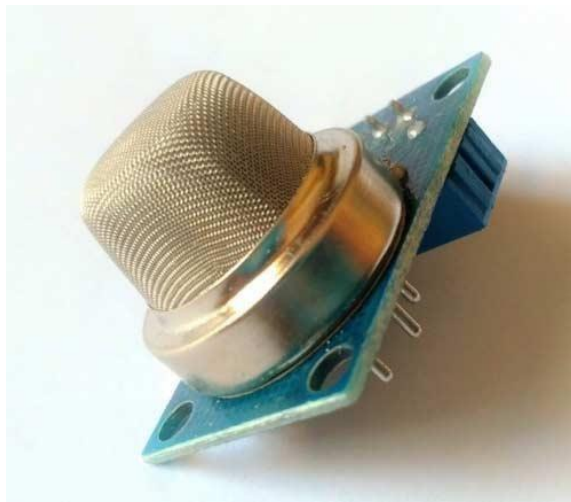
Gyroscope sensor to sense and determine the orientation with the help of Earth's gravity i.e. it measures the angular velocity. The main difference between Accelerometers & Gyroscope Sensors is that Gyroscope can sense the rotation where accelerometer can't. In other words, Gyroscope measure any rotation and unaffected by acceleration and Accelerometer cannot distinguish rotation from acceleration and unable to work when in center of rotation.



1.6] Gas sensor:

In industrial applications gas sensors play a major role in detecting the gas leakage. If no such device is installed in such areas it ultimately leads to an unbelievable disaster. These gas sensors are classified into various types based on the type of gas that is to be detected. Let's see how this sensor works.

Underneath a metal sheet there exists a sensing element which is connected to the terminals where a current is applied to it. When the gas particles hit the sensing element, it leads to a chemical reaction such that the resistance of the element varies and current through it also alters which finally can detect the gas.



1.7] Hall Sensor:

The name itself states that it is the sensor which works on the Hall Effect. It can be defined as when a magnetic field is brought close to the current carrying conductor (perpendicular to the direction of the electric field) then a potential difference is developed across the given conductor. Using this property a *Hall sensor is used to detect the magnetic field* and gives output in terms of voltage. Care should be taken that the Hall sensor can detect only one pole of the magnet.



The hall sensor is used in few smartphones which are helpful in turning off the screen when the flap cover (which has a magnet in it) is closed onto the screen. Here is one practical application of [Hall Effect sensor in Door Alarm](#).

1.8] **Flex Sensor:**

A FLEX sensor is a transducer which *changes its resistance when its shape is changed or when it is bent*. A FLEX sensor is 2.2 inches long or of finger length. It is shown in the figure. Simply speaking the sensor terminal resistance increases when it's bent. This change in resistance can do no good unless we can read them. The controller at hand can only read the changes in voltage and nothing less, for this, we are going to use [voltage divider circuit](#), with that we can derive the resistance change as a voltage change. Learn here about [how to use Flex Sensor](#).



1.9] **Microphone (Sound Sensor):**

Microphone can be seen on all the smartphones or mobiles. It can detect the audio signal and convert them into small voltage (mV) electrical signals. A microphone can be of many types like condenser microphone, crystal

microphone, carbon microphone etc. each type of microphone work on the properties like capacitance, piezoelectric effect, resistance respectively. Let us see the operation of a crystal microphone which works on the piezoelectric effect. A bimorph crystal is used which under pressure or vibrations produces proportional alternating voltage. A diaphragm is connected to the crystal through a drive pin such that when the sound signal hits the diaphragm it moves to and fro, this movement changes the position of the drive pin which causes vibrations in the crystal thus an alternating voltage is generated with respect to the applied sound signal. The obtained voltage is fed to an [amplifier](#) in order to increase the overall strength of the signal. Here are various [circuits based on Microphone](#).



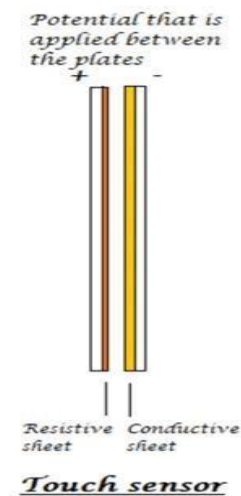
1.10] Touch Sensor:

In this generation, we can say that almost all are using smartphones which have widescreen that too a screen which can sense our touch. So, let's see how this touchscreen works. Basically, there are two types of touch sensors *resistive based and a capacitive based touch screens*. Let's know about working of these sensors briefly.

The *resistive touchscreen* has a resistive sheet at the base and a conductive sheet under the screen both of these are separated by an air gap with a small voltage applied to the sheets. When we press or touch the screen the conductive sheet touches the resistive sheet at that point causing current flow at that particular point, the software senses the location and relevant action is performed.

Whereas *capacitive touch* works on the electrostatic charge that is available on our body. The screen is already charged with the all electric field. When we touch the screen a close circuit forms due to electrostatic charge that flow through our body. Further, software decides the location and the action to be performed. We can observe that capacitive touch screen won't work when

wear hand gloves because there won't be conduction between the finger(s) and the screen.



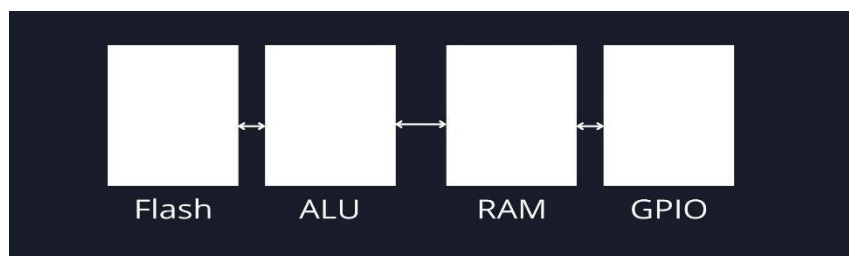
2] Description of various Controllers are as follows:

2.1] **Arduino:**

To understand it we need to understand 8-bit microcontrollers because an Arduino is a nothing but an ATmega328p microcontroller.

What does an Arduino (ATMEGA328P) consist of internally?

- Flash
- ALU
- RAM
- GPIO



Arduino Architecture

What is Flash memory in Arduino (ATMEGA328P)?

A flash is a memory type like RAM, but it's a much faster one. It stores the Arduino code and then sends constructions to ALU to do operations, store bits etc.

What is Arduino (ATMEGA328P) ALU?

ALU is the main unit responsible for most Arithmetic Operations like Addition, subtraction, multiplication, division, memory addressing operations and many more. It consists of transistors connected in certain topologies to achieve those calculation goals

What is Arduino (ATMEGA328P) RAM unit?

Memory is where program calculations results, variables initialization and declaration is stored. The more RAM memory exists, the more sophisticated programs can be written.

What is Arduino (ATMEGA328P) GPIO?

GPIOs are the pin header you see in the Arduino picture above labelled in red, by using them we can read from sensors, activate motors and interact with the outer world.



2.2] Raspberry Pi:

- Is much more powerful in term of CPU, and even though it has the same fixed architecture like other microcontroller like Arduino, Raspberry PI is more powerful because it contains a GPU which makes it able to run an OS (operating system).
- Is used as a mini full scale processing unit.
- Has WIFI module so that you can connect it to your router and wireless devices.
- Much powerful than Arduino and has a much larger RAM.



2.3] Node MCU (ESP8266):

- Is very similar to the Arduino architecture (It differently has a built in WIFI)
- More powerful than Arduino in terms of flash
- Mainly used in IoT projects.
- Less powerful than Raspberry Pi.
- Can simply execute your program.



3] Description of various types of wireless modules:

Wireless communication module:

Wireless communication modules are widely used in vehicle monitoring, remote control, telemetry, small wireless networks, wireless meter reading, access control systems, community paging, industrial data acquisition systems, wireless tags, identification, noncontact RF smart cards, small wireless data terminals, security Fire protection system, wireless remote control system, biological signal acquisition, hydrometeorological monitoring, robot control, wireless 232 data communication, wireless 485/422 data communication, digital audio, digital image transmission, and other fields.

The common wireless communication module:

3.1] Bluetooth:

Bluetooth is a wireless communication module that we often come into contact with every day. This technology was invented by the then telecommunications

giant Ericsson in 1994. It is a wireless technology standard that can realize fixed terminal equipment, mobile terminal equipment, and personal area networks. For short-distance data exchange, use UHF radio waves in the 2.4~2.485GHZ ISM band.

The complexity of Bluetooth wireless technology is relatively high; the device networking speed is fast, only 10 seconds; the integration and reliability are very high; the transmission rate is generally 1Mbps; the cost is low, and the installation is relatively simple. It is a suitable short-range wireless communication technology.

3.2] Wi-Fi:

Wi-Fi wireless technology is very attributed to everyone, and it is the most wireless communication technology we use daily. It is a wireless LAN technology created with the IEEE802.11 standard. This technology converts all wired network signals into radio wave signals, and other terminal devices connect to Wi-Fi through a wireless communication module to realize wireless network communication.

The coverage of Wi-Fi technology is generally within 100 meters, the technology is more complicated, the transmission rate can reach 54Mbps, the working frequency band is 2.4GHZ, and the transmission power is less than 100mw. Compared with Bluetooth wireless communication, data security performance is poor. However, the invention of Wi-Fi is in line with modern personal and social needs, and its development prospects are very huge.

3.3] ZigBee:

ZigBee wireless communication technology is a low-power local area network protocol based on the IEEE802.15.4 standard. It was formally established in August 2001. At the beginning of its establishment, due to the hasty release of the version, there were certain errors, and the subsequent development continued to improve.

ZigBee wireless communication technology is similar to Bluetooth wireless communication technology, both are short-range wireless communication technologies, but Bluetooth wireless communication technology has many shortcomings, such as high power consumption, high complexity, short communication distance, etc., which is only suitable for family and personal use. The development of ZigBee technology is to meet the needs of industrial

automation. It has a simple layout, anti-interference, reliable transmission, convenient use, and low cost. The communication distance ranges from 10 meters in Bluetooth to hundreds of meters in open spaces and about 50 meters indoors.

3.4] Traditional digital radio:

Digital radio station refers to a high-performance professional data transmission radio station with the help of DSP digital signal processing technology and software wireless point technology. Digital radio can be understood as a communication medium, which has a certain range of use like optical fiber and microwave. The transmission distance of the digital radio station is very long, suitable for various complicated environments, and the transmission rate is 19.2Kbps, but the terminal equipment is more expensive, the use cost is also high, and the installation is more complicated.

3.5] GSM:

GSM is a globally recognised digital cellular communication protocol. The GSM standard was developed by the European Telecommunications Standards Institute to describe the procedures for second-generation digital mobile networks, such as those used by mobile phones. It is a broad-area communications technology programme that uses digital radio channelling to provide audio, information, and multimedia communication systems. Every GSM radio channel is 200 kHz broad and is further divided into frames of eight time slots. The GSM system consists of mobile stations, base stations, and interweaving switching systems.

The GSM programme allows 8 to 16 audio users to share a single radio channel, and each radio transmission station can have numerous radio channels. Because of its simplicity, cost, and accessibility, GSM is now the most often utilised network technology in the Internet of Things (IoT).

